

## Sets

In general sets will be denoted by capital letters:  $A, B, C, \dots$

Frequently used sets:

$\emptyset$  = the empty set (i.e. the set that contains no elements)

$\mathbb{N} = \{0, 1, 2, \dots\}$  the set of natural numbers

$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$  the set of integers

$\mathbb{Z}^+ = \{1, 2, 3, \dots\}$  the set of positive integers

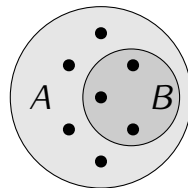
$\mathbb{Q}$  = the set of rational numbers

$\mathbb{R}$  = the set of real numbers

We will write  $x \in A$  to denote that  $x$  is an element of the set  $A$  and  $y \notin A$  to indicate that  $y$  is not an element of  $A$ . For example,  $5 \in \mathbb{Z}$ ,  $\frac{1}{3} \notin \mathbb{Z}$ .

## Subsets

A set  $B$  is a *subset* of a set  $A$  if every element of  $B$  is in  $A$ . In such case we write  $B \subseteq A$ .



A set  $B$  is a *proper subset* of  $A$  if  $B \subseteq A$  and  $B \neq A$ .

**Example.** Some subsets of  $\mathbb{Z}$ :

- $A = \{n \in \mathbb{Z} \mid n > 10\}$  - the set of integers greater than 10.
- $B = \{n \in \mathbb{Z} \mid n = 2k \text{ for some } k \in \mathbb{Z}\}$  - the set of even integers.

## Operations on sets

- The *union* of sets  $A$  and  $B$  is the set  $A \cup B$  that consists of all elements that belong to either  $A$  or  $B$  (or both):

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

- The *intersection* of sets  $A$  and  $B$  is the set  $A \cap B$  that consists of all elements that belong to both  $A$  and  $B$ :

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

- The *difference* of sets  $A$  and  $B$  is the set  $A \setminus B$  that consists of all elements that belong to  $A$  but not to  $B$ :

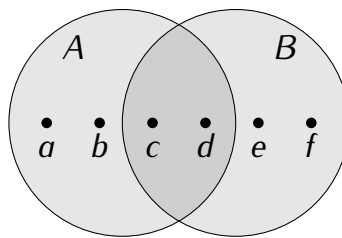
$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$$

**Example.** If  $A = \{a, b, c, d\}$ ,  $B = \{c, d, e, f\}$  then:

$$A \cup B = \{a, b, c, d, e, f\}$$

$$A \cap B = \{c, d\}$$

$$A \setminus B = \{a, b\}$$



**Note.** We say that sets  $A$  and  $B$  are *disjoint sets* if  $A \cap B = \emptyset$ .

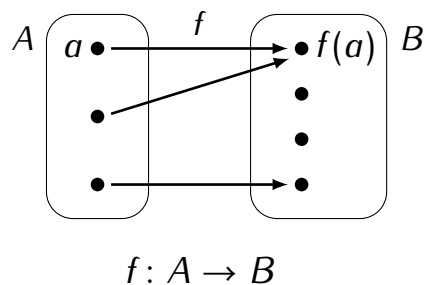
- The *Cartesian product* of sets  $A$ ,  $B$  is the set consisting of all ordered pairs of elements of  $A$  and  $B$ :

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

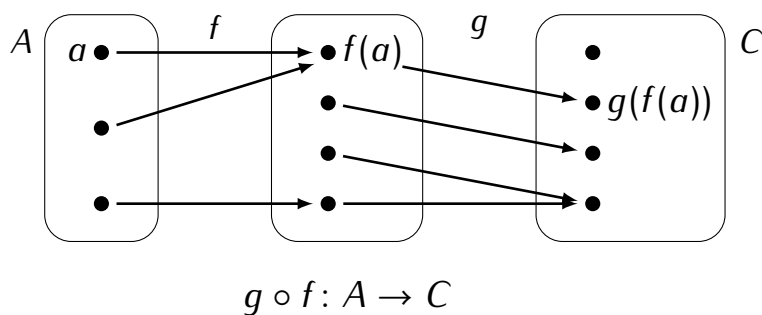
**Example.** If  $A = \{1, 2, 3\}$ ,  $B = \{2, 3, 4\}$  then

$$A \times B = \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$$

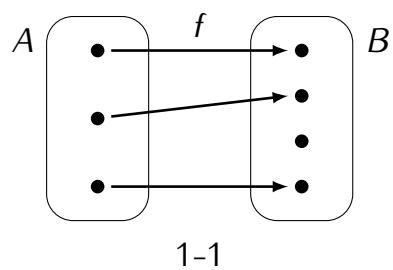
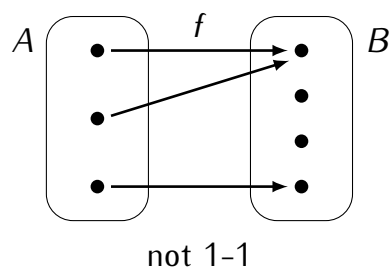
## Functions



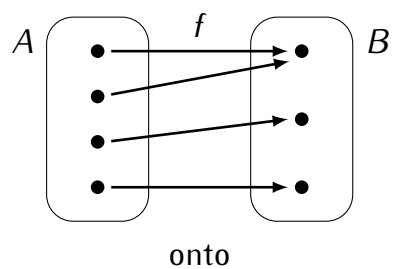
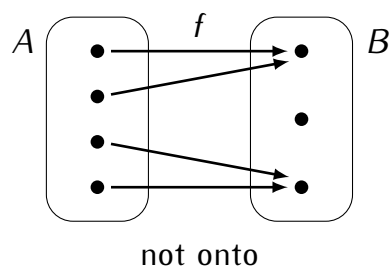
Composition of functions:



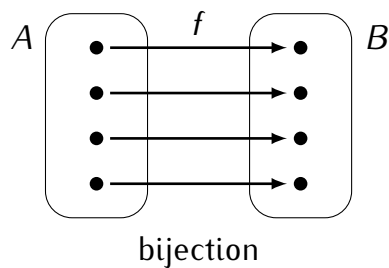
- A function  $f: A \rightarrow B$  is *1-1* if  $f(a) = f(a')$  only if  $a = a'$ .



- A function  $f: A \rightarrow B$  is *onto* if for every  $b \in B$  there is  $a \in A$  such that  $f(a) = b$



- A function  $f: A \rightarrow B$  is a *bijection* if  $f$  is both 1-1 and onto.



**Note.** If  $f: A \rightarrow B$  is a bijection then the inverse function  $f^{-1}: B \rightarrow A$  exists and it is also a bijection. Then for every  $a \in A$  and  $b \in B$  we have:

$$f^{-1}(f(a)) = a \quad \text{and} \quad f(f^{-1}(b)) = b$$

### Cardinality

We will denote by  $|A|$  the cardinality of the set  $A$ . For a finite set, this is the number of elements of  $A$ .

**Note.** If  $A, B$  are sets, then  $|A| = |B|$  if and only if there exists a bijection  $f: A \rightarrow B$ .